MILITARY SPECIFICATION

FOR
PROJECTILE, 81MM SMOKE, RP, M819
METAL PARTS

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the metal parts used on one type of 81MM Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315
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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS (See 6.3)

9327839	-	Cartridge, 81MM Smoke, RP, M819 with Fuze, Time, M772

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	-	Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Method
E-8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications. Materials shall be free of defects such as cracks, splits, porosity, shrinkage, seams, cold shuts, pipe-type defects, planar defects, inclusions or similar defects.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9327839 and associated drawings, and with all requirements specified in applicable specifications and standards. Parts shall be free of burrs which may interfere with further assembly or proper functioning.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2).

3.4 Workmanship.

3.4.1 Metal characteristics. All components shall be free from gouges, dents, cracks, deformation or any similar defects.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

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3.4.3 Burrs. No part shall have a burr which might be injurious to personnel handling the item.

3.4.4 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease and other foreign matter. The cleaning and deburring method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

3.4.5 Markings. Markings shall be neat, complete and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements in sections 3 and 5. The inspections set forth in th specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample

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shall consist of the following items in sample quantities as indicated (See Table I).

Part description	Drawings	Quantity
Shear Pin	9300226	10
Piston plate	9327829	5
Adapter	9327835	5
Body tube	9327837	5
Rear support disc	9349482	5
Tail cone	9349624	5
Flat-type tensile test specimens (2 tail cone, 2 support disc)		4
"T"-type tensile test specimens (body tube)		2

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

MIL-P-70551

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, 81MM Smoke, RP, M819, Metal Parts for	1 of 1	See below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Shear pin (dwg. 9300226)				
	Examination for Defects	10	100%	3.2	4.4.2.3
	Shear strength test	10	100%	3.2	4.5.3
	Piston Plate (dwg. 9327829)				
	Examination for Defects	5	100%	3.2	4.4.2.7
	Adapter (Dwg. 9327835)				
	Examination for Defects	5	100%	3.2	4.4.2.4
	Body Tube (Dwg. 9327837)				
	Examination for Defects	5	100%	3.2	4.4.2.5
	Tensile Test Specimens	2	100%	3.2	4.5.2
	Rear Support Disc (Dwg. 9349482)				
	Examination for Defects	5	100%	3.2	4.4.2.8
	Tensile Test Specimens	2	100%	3.2	4.5.2
	Tail Cone				
	Examination for Defects	5	100%	3.2	4.4.2.2
	Tensile Test Specimens	2	100%	3.2	4.5.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of metal parts shall contain:

a. Shear pins of one lot interfix number, from one manufacturer.

b. Piston plates of one lot interfix number, from one manufacturer.

c. Adapters of one lot interfix number, from one manufacturer.

d. Body tubes of one lot interfix number from one manufacturer.

e. Rear support discs interfix number from one manufacturer.

f. Tail cones from one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
4.4.2.1	Tensile Test Specimens				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tension test (body tube)	2 1/	100%	3.1	4.5.2.1
102	Tension test (tail cone)	2 1/	100%	3.2	4.5.2.2
103	Tension test (rear support disc)	2 1/	100%	3.1	4.5.3.3

NOTES:
1/ Failure of any sample shall cause the applicable material interfix to be rejected.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.2	Cone, Tail				9349624
					NEXT HIGHER ASSEMBLY 9300233
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of rear thread with rear face and forward outside diameter (O.D.) except pilot diameter (dia.)		0.40%	3.2	Gage
102	Perpendicularity of rear face with pitch diameter (P.D.) of rear thread		0.40%	3.2	Gage
103	Runout at rear of taper with forward O.D. except pilot dia.		0.40%	3.2	Gage
104	Runout at front of taper with forward O.D. except pilot dia.		0.40%	3.2	Gage
105	Depth of interlock		0.40%	3.2	Gage
106	Runout of obturator groove with bourrelet (large O.D.)		0.40%	3.2	Gage
107	Runout of bourrelet with forward O.D. except pilot dia.		0.40%	3.2	Gage
108	Runout of forward inside diameter (I.D.) with forward O.D. except pilot dia.		0.40%	3.2	Gage
109	Wall thickness		0.40%	3.2	Gage
110	Obturator groove diameter		0.40%	3.2	Gage
111	Bourrelet diameter		0.40%	3.2	Gage
112	Pitch diameter of forward thread		0.40%	3.2	Gage
113	Minor diameter of forward thread		0.40%	3.2	Gage
114	Pitch diameter of rear thread		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.2	Cone, Tail (Continued)		9349624
			NEXT HIGHER ASSEMBLY 9300233

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Minor diameter of rear thread, max.		100%	3.2	Gage
Major					
116	Distance from front face to start of interlock		0.40%	3.2	Gage
117	O.D. of interlock joint		0.40%	3.2	Gage
118	O.D. forward of interlock joint		0.40%	3.2	Gage
119	Overall length		0.40%	3.2	Gage
120	Distance from front face to obturator shoulder		0.40%	3.2	Gage
121	I.D. rear of front threads		0.40%	3.2	Gage
122	Distance from rear face to base of cavity		0.40%	3.2	Gage
123	Width of obturator groove		0.40%	3.2	Gage
124	Tensile test	2	100%	3.2	Gage/4.5.2.2
125	Runout of forward thread P.D. with forward O.D. except pilot dia.		0.40%	3.2	Gage
Minor					
201	Length from front face to rear taper diameter		0.65%	3.2	Gage
202	Distance from front face to forward taper diameter		0.65%	3.2	Gage
203	Surface finish improper		0.65%	3.2	Gage
204	Protective finish improper		0.65%	3.2	Visual
205	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (See 6.6)

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cone, Tail (Continued)	3 OF 3	9349624
			NEXT HIGHER ASSEMBLY 9300233

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
206	Pilot diameter missing		0.65%	3.2	Visual
207	Presence of paint on forward O.D.		0.65%	3.2	Visual
208	Depth of forward thread		0.65%	3.2	Gage
209	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9300226
4.4.2.3	Pin, Shear				NEXT HIGHER ASSEMBLY 9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter incorrect		0.40%	3.2	Gage
102	Excessive chamfer		0.40%	3.2	Visual (See 6.6)
103	Shear strength	1/10	100%	3.2	Gage/4.5.3
Minor					
201	Chamfer missing or inadequate		0.65%	3.2	Visual (See 6.6)
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

1/ One pin at a time to be tested.

AMSC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Adapter				9327835
					NEXT HIGHER ASSEMBLY
					9327832
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward outside diameter		0.40%	3.2	Gage
102	Rear outside diameter		0.40%	3.2	Gage
103	Runout of forward O.D. with rear O.D. and front shoulder		0.40%	3.2	Gage
104	Perpendicularity of shoulder to rear O.D.		0.40%	3.2	Gage
105	Parallelism of front face with shoulder		0.40%	3.2	Gage
106	Pitch diameter of threads		0.40%	3.2	Gage
107	Minor diameter of threads, max.		0.40%	3.2	Plug gage
108	Metal defective		0.40%	3.1	Visual
109	Distance from front face to shoulder		0.40%	3.2	Gage
110	Overall length		0.40%	3.2	Gage
Minor					
201	Diameter at front of thread chamfer		0.65%	3.2	Gage
202	Diameter at rear of rear outside chamfer		0.65%	3.2	Gage
203	Radii or chamfers missing or incorrect		0.65%	3.2	Visual (See 6.6)
204	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Tube, Body				9327837
					NEXT HIGHER ASSEMBLY 9327832
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parallelism of front face with rear face		0.40%	3.2	Gage
102	Forward I.D. (fuze adapter cavity)		0.40%	3.2	Gage
103	Forward O.D.		0.40%	3.2	Gage
104	Central I.D.		0.40%	3.2	Gage
105	Rear I.D. (tail cone cavity)		0.40%	3.2	Gage
106	O.D. forward of rear shoulder		0.40%	3.2	Gage
107	Thickness of wall, min.		0.40%	3.2	Gage
108	Runout of rear O.D. with rear I.D. and rear face		0.40%	3.2	Gage
109	Runout of forward O.D. with rear face and rear I.D.		0.40%	3.2	Gage
110	Perpendicularity of rear face with rear I.D.		0.40%	3.2	Gage
111	Metal defective		0.40%	3.1	Visual
112	Length, total		0.40%	3.2	Gage
113	Depth of tail cone cavity (rear I.D.)		0.40%	3.2	Gage
114	Depth from rear face to interlock shoulder		0.40%	3.2	Gage
115	Rear O.D. (interlock joint)		0.40%	3.2	Gage
116	Depth of fuze adapter cavity (forward I.D.)		0.40%	3.2	Gage
117	Length of bourrelet (forward O.D.)		0.40%	3.2	Gage
118	Tensile test	2	100%	3.1	Gage/4.5.2.1

NOTES:

AMSMC (m 1570, 1 Feb 85 Repl(s DRSMC-QA (D) Form 160, 1 Aug 83, which may not(? used.

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	Tube, Body (Continued)				9327837
					NEXT HIGHER ASSEMBLY 9327832
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Protective finish improper		0.65%	3.2	Visual
202	Radii or chamfers missing or improper		0.65%	3.2	Visual (see 6.6)
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Body Tube Assembly				9327832
					NEXT HIGHER ASSEMBLY 9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of paint on rear I.D. or forward threads		0.40%	3.2	Visual
102	Forward O.D., max.		0.40%	3.2	Gage
103	True position of forward O.D. with rear I.D. and front and rear faces		0.40%	3.2	Gage
104	Retaining pin cavity diameter (6 places)	1/		3.2	Gage
105	Retaining pin cavity depth (6 places)	1/		3.	Gage
106	Absence or looseness of retaining pins		0.40%	3.2	Visual/Tactile
107	Distance from front face to retaining pin holes (6 places)	1/		3.2	Gage
108	Gap between body tube and fuze adapter, max.		0.40%	3.2	Gage
109	Length, max.		0.40%	3.2	Gage
110	Retaining pin(s) above flush or more than .015 below flush		0.40%	3.2	Gage
111	Presence of burrs in threads		0.40%	3.2	Visual
Minor					
201	Marking missing, incorrect, illegible or improperly located		0.65%	3.2	Visual
202	Radii or chamfer missing or incorrect		0.65%	3.2	Visual (See 6.6)
203	Finish improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES: 1/ At the start of each shift 10 assemblies shall be drilled but the retaining pins shall be omitted. Failure of any hole to meet the applicable requirement shall require adjustment of tooling.

AMSMC rm 1570, 1 Feb 85 Rep es DRSMC-QA (D) Form 160, 1 Aug 83, which may n e used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Plate, Piston				9327829
					NEXT HIGHER ASSEMBLY 9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Hole missing		0.40%	3.2	Visual
102	Diameter of plate		0.40%	3.2	Gage
103	Thickness of plate		0.40%	3.2	Gage
104	Metal defective		0.40%	3.1	Visual
105	Diameter of hole		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.8	Disc, Rear Support				9349482
					NEXT HIGHER ASSEMBLY 9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.1	Visual
Major					
101	Large O.D.		0.40%	3.2	Gage
102	OD forward of threads		0.40%	3.2	Gage
103	OD rear of threads		0.40%	3.2	Gage
104	Length of undercut		0.40%	3.2	Gage
105	Length, overall		0.40%	3.2	Gage
106	Depth of well at rear		0.40%	3.2	Gage
107	Inside diameter of well		0.40%	3.2	Gage
108	Pitch diameter (PD) of threads		0.40%	3.2	Gage
109	Major diameter of threads		0.40%	3.2	Gage
110	Perpendicularity of thread PD to shoulder behind front face		0.40%	3.2	Gage
111	Distance from rear face to shoulder		0.40%	3.5	Visual
112	Tensile test	2	100%	3.1	Gage/4.5.2.3

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70551 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Disc, Rear Support (Continued)				9349482
					NEXT HIGHER ASSEMBLY
					9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of spanner holes		0.65%	3.2	Gage
202	Depth of spanner holes		0.65%	3.2	Gage
203	Distance between spanner holes		0.65%	3.2	Gage
204	Spanner hole(s) missing		0.65%	3.2	Visual
205	Length of rear OD		0.65%	3.2	Gage
206	Radii missing or improper		0.65%	3.2	Visual/See 6.6
207	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Tension test.

4.5.2.1 Body tube. From each of two tubes selected from one interfix, one tension test specimen shall be prepared. The samples shall be the longest standard size "T" type specimen conforming to ASTM E-8 which can be taken from the constant wall thickness portion of body tube. Failure of any specimen to meet applicable requirements shall be cause for rejection of the applicable material interfix.

4.5.2.2 Tail cone. From each of two tail cones selected from one interfix, one tension test specimen of the longest obtainable flat specimen conforming to ASTM E-8 shall be taken longitudinally from the tapered portion of the component. Failure of any specimen to meet applicable requirements shall be cause for rejection of the applicable material interfix.

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4.5.2.3 Support disc. From each of two support discs selected from one interfix, one tension test specimen of the longest obtainable flat-type conforming to ASTM E-8 shall be taken from the solid section between the two spanner holes. Failure of any specimen to meet applicable requirements shall be cause for rejection of the applicable material interfix.

4.5.3 Shear strength test. Shear pins from one shear pin lot shall be tested in approved fixturing, one at a time. Failure of any pin to meet the requirement specified on the applicable drawing shall cause rejection of the shear pin lot.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing, and marking of metal parts shall be in accordance with the requirements of level C in MIL-STD-1169.

6. NOTES

6.1 Intended use. The components covered by this plan are intended for assembly to M819 smoke cartridges for use on the 81MM mortar M252.

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal and US Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.4 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-I(D), Dover, NJ 07801-5001.

6.5 Alternate procedures. Alternate procedures to those contained in this specification shall be approved in advance by: Commander, USA ARDEC, ATTN: AMSMC-QAT-B (D), Dover, NJ 07801-5001.

6.6 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, these characteristics may be measured or gaged to determine acceptability.

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6.7 Surface roughness. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

6.8 Subject term (key word) listing.

Ammunition, mortar
Cone, tail
Disc, support
Pin, shear
Specimen, tensile
Tube, body

Custodian:
ARMY-AR

Preparing activity:
ARMY-AR

(Project - 1315-A681)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70551	PROJECTILE, 81MM SMOKE, RP, M819 MATAL PARTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include A Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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